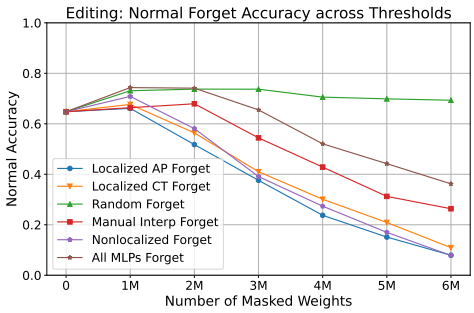

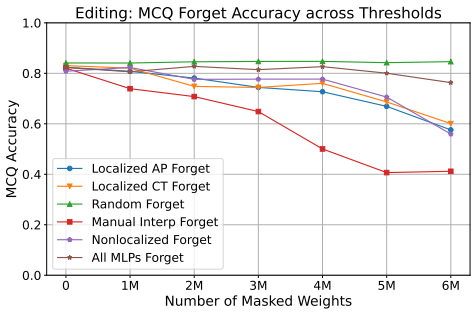

Figure 21: Editing CounterFact facts. **(Left)** Testing models accuracy on the normal forget set. **(Right)** Testing the models' accuracy in the MCQ prompt format.

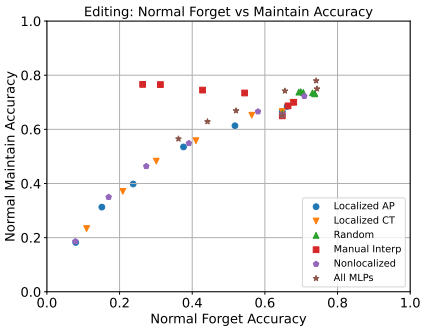

Figure 22: Accuracy on normal forget set vs on the maintain set across localizations and discretization thresholds.

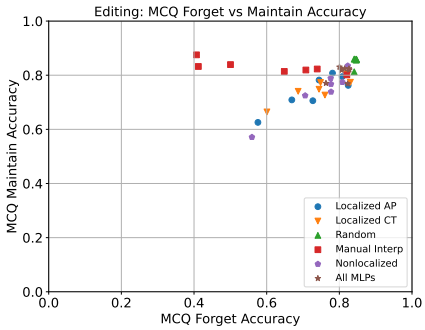

Figure 23: Accuracy on multiple choice input vs on the maintain set across localizations and discretization thresholds.

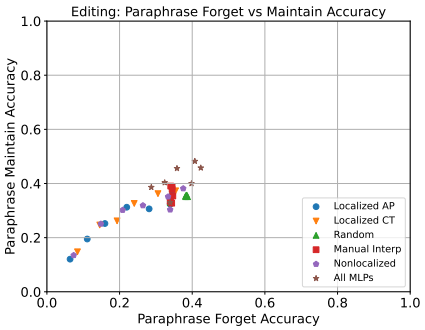

Figure 24: Accuracy on paraphrased input vs on the maintain set across localizations and discretization thresholds.

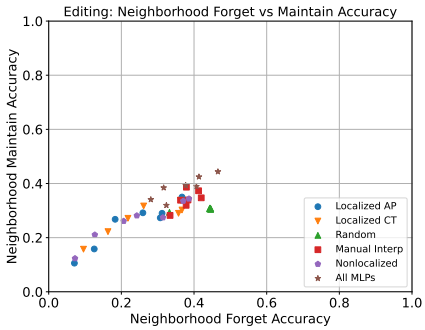

Figure 25: Accuracy on "neighborhood" input vs on the maintain set across localizations and discretization thresholds.

A.6 Hyperparameters

Across all tasks except Sequential-CounterFact-Editing and all models, we fine tune using 50 iterations of batch size 4 with 16 accumulation steps, using an AdamW optimizer (Kingma & Ba, 2017) with 0 weight decay and a cosine annealing scheduler. For Gemma-2-9b, we are forced to use an 8-bit optimizer to fit our training in the memory of 1 GPU. We find that the optimal learning rate is quite sensitive to the localization used and the edit task, so we first sweep over learning rates to find reasonable learning rates. We sweep over the learning rates of 2e-6, 5e-6, 1e-5, 2e-5, 5e-5, and 1e-4, training models over 50 iterations with all λs set to 1. We also tune the λ_1 parameter associated with the $L_{\text{injection}}$ cross entropy loss. We don't tune the other λ parameters because they are all

Table 2: Comparison of total parameters of each mechanism that are present in each localization, for editing 16 facts from CounterFact

Localization	Extraction Heads	Extraction MLPs	Fact Lookup
Total	27,448,320	1,027,604,480	1,130,364,928
Attrib. Patching	13,724,160 (50.0%)	616,562,688 (60.0%)	102,760,448 (9.1%)
Causal Tracing	8,234,496 (30.0%)	308,281,344 (30.0%)	411,041,792 (36.4%)
FLU	0	0	1,130,364,928 (100.0%)
All-MLPs	0	1,027,604,480 (100.0%)	1,130,364,928 (100.0%)
Nonlocalized	27,448,320 (100.0%)	1,027,604,480 (100.0%)	1,130,364,928 (100.0%)

Table 3: Comparison of parameters of each mechanism that are masked by a trained weight mask, discretized to about 6 million weights

Localization Type	Extraction Heads	Extraction MLPs	Fact Lookup
Total (baseline)	27,448,320 (100%)	1,027,604,480 (100%)	1,130,364,928 (100%)
Attrib. Patching	165,300 (0.60%)	1,479,877 (0.14%)	1,385,198 (0.12%)
Causal Tracing	30,828 (0.11%)	1,491,040 (0.15%)	1,424,059 (0.13%)
FLU	0 (0.0%)	0 (0.0%)	6,248,039 (0.55%)
All-MLPs	0 (0.0%)	1,378,744 (0.13%)	1,663,772 (0.15%)
Nonlocalized	358,918 (1.3%)	1,198,211 (0.12%)	1,174,939 (0.10%)

maintenance losses, and setting them to 1 works sufficiently to maintain performance across almost all setups.

For the Sequential-CounterFact-Editing task, we use the same hyperparameters from the CounterFact-Editing task and we train for 100 total iterations rather than 50, using 25 for each subset of 16 facts. We choose 25 iterations because it balances between being half the number of iterations we typically use per subset of that size, and also double the number of steps overall as we use in CounterFact-Editing.

To avoid leaking evaluation information through this sweep process, we optimize learning rate for the objective of $(1 - \textit{Forget Set Ground Truth Accuracy}) + \textit{Forget Set Edit Accuracy} + \textit{Maintain Set Ground Truth Accuracy} + \textit{Pile Accuracy}$, avoiding any of our robustness metrics (one could view this sweep process as simply another part of the training process, since we only use train-time information). We run sweeps for Causal Tracing, Manual Fact Lookup, and No Localization. We then use the hyperparameters from Causal Tracing for Attribution Patching and Random (which all localize to MLPs and attention components), we use the hyperparameters from Fact Lookup for Random MLPs, Causal Tracing MLPs, and Attribution Patching MLPs (all localize to the same number of MLPs), and we use the hyperparameters from No Localization for All MLPs (which have the largest number of active parameters).

A.6.1 Sample Hyperparameter Sweep

In Table 4, we show the full results of one sweep, optimizing learning rate for editing 16 facts from CounterFact. Especially for No Localization, some learning rates fail to edit in the correct answer with high accuracy, or fail to maintain accuracy on the maintain set. In Table 5, we see that editing results are not particularly sensitive to the coefficient used with the injection cross entropy loss.

A.6.2 All Hyperparameters Used

Table 6 has all learning rates used and Table 7 has all injection loss coefficients used.

A.7 Evaluation Details

A.7.1 Details on Standard Prompt Evaluations

We report standard metrics of Forget Error, Edit Accuracy, and Maintain Accuracy, in the same prompt format that the models were trained on. These metrics are optimized by the loss, so we

Table 4: Gemma-7b learning rate sweep, editing 16 CounterFact facts.

	Pile Accuracy ↑	Forget Accuracy ↓	Edit Accuracy ↑	Maintained Accuracy ↑	Overall Score ↑
FLU					
LR 0.0001	0.488	0.000	1.000	0.698	3.186
LR 1e-05	0.513	0.000	1.000	0.975	3.488
LR 2e-05	0.542	0.000	1.000	0.950	3.492
LR 2e-06	0.499	0.007	0.961	0.869	3.323
LR 5e-05	0.520	0.000	1.000	0.822	3.342
LR 5e-06	0.528	0.000	0.999	0.980	**3.507**
Localized CT					
LR 0.0001	0.462	0.001	0.999	0.697	3.157
LR 1e-05	0.513	0.000	1.000	0.984	3.496
LR 2e-05	0.507	0.000	1.000	0.961	3.467
LR 2e-06	0.540	0.036	0.921	0.849	3.274
LR 5e-05	0.488	0.000	1.000	0.846	3.334
LR 5e-06	0.537	0.000	1.000	0.982	**3.519**
Nonlocalized					
LR 0.0001	0.062	0.032	0.557	0.094	1.680
LR 1e-05	0.520	0.000	1.000	0.892	3.412
LR 2e-05	0.479	0.001	0.998	0.807	3.284
LR 2e-06	0.529	0.000	1.000	0.982	3.510
LR 5e-05	0.046	0.034	0.710	0.092	1.815
LR 5e-06	0.536	0.000	1.000	0.988	**3.524**

Table 5: Gemma-7b inject loss coefficient sweep, editing 16 CounterFact facts.

	Pile Accuracy ↑	Forget Accuracy ↓	Edit Accuracy ↑	Maintained Accuracy ↑	Overall Score ↑
FLU					
FC 0.1	0.538	0.000	0.998	0.990	**3.525**
FC 0.2	0.510	0.000	0.999	0.981	3.491
FC 0.5	0.508	0.000	0.999	0.985	3.493
FC 1	0.510	0.000	0.999	0.973	3.483
FC 2	0.532	0.000	1.000	0.984	3.515
FC 5	0.534	0.000	1.000	0.985	3.518
Localized CT					
FC 0.1	0.532	0.004	0.995	0.983	3.506
FC 0.2	0.535	0.001	0.998	0.986	**3.518**
FC 0.5	0.524	0.000	0.999	0.974	3.497
FC 1	0.519	0.000	1.000	0.988	3.507
FC 2	0.536	0.000	1.000	0.979	3.514
FC 5	0.518	0.000	1.000	0.975	3.493
Nonlocalized					
FC 0.1	0.525	0.001	0.998	0.969	3.492
FC 0.2	0.562	0.000	0.999	0.982	**3.543**
FC 0.5	0.524	0.000	1.000	0.962	3.486
FC 1	0.531	0.000	1.000	0.980	3.511
FC 2	0.514	0.000	1.000	0.978	3.492
FC 5	0.528	0.000	1.000	0.975	3.503

expect all localizations to do almost perfectly on these evaluations. Figures 26 to 29 show that localizations perform approximately equivalently on these basic evaluations across all tasks.

A.7.2 Details on Adversarial Prompt Evaluations

We report the adversarial prompt evaluations from Section 3.1 across all localizations. Figures 30 to 33 all show that FLU localization is more robust in MCQ compared to every other localization (significantly stronger for CounterFact). Figures 32 and 33 show that FLU localization is optimal in Paraphrase and Neighborhood in all cases except for Paraphrase compared to the Random localization in Sequential-CounterFact-Editing.